

Tara Dixit

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EDUCATION

Stanford University — B.S. in Computer Science, AI Track

Graduation Date – Jun 2027

EXPERIENCE

AI Safety & Security Research Intern

Stealth Startup | Jul 2025 – Present

- Evaluated AI agents across 120 runs on 15 Linux setup and debugging tasks, measuring safe command use, verification, unnecessary system changes, and recovery from failures.
- Investigated guardrail DoS and cyber-agent failure modes, and designed behavioral logs to capture agent actions, policy-relevant behavior, and recovery after failures.

CURIS Fellow

Stanford University | Jun 2025 – Mar 2026

- Analyzed 31 threat reports from incident-response firms and cyber insurers to compare how they describe real-world attacks and prioritize threats.
- Found major disagreement on initial access and exploited vulnerabilities, which limits the reports' usefulness for security teams.

Research Assistant — Empirical Security Research Group

Stanford University | Jan 2025 – Jun 2025

- Contributed to a USENIX Security 2025 study of cybersecurity guidance across 41 countries.
- Built a comparison taxonomy and used LLMs to compare recommendations, finding inconsistencies that could create false confidence about security coverage.

PROJECTS

ContainmentBench — Security Boundaries for Coding Agents

Jun 2026 – Present

- Built a Linux testbed for evaluating coding agents under file, process, and network access constraints during multi-step tasks.
- Logged agent runs to detect and trace attempted policy violations across security boundaries.

Hidden Reliability Failures in Malware Detection

Mar 2026 – Aug 2026

- Evaluated the EMBER2024 malware detector on 540,000 Windows executable files.
- Found that 98.0% overall accuracy and strong aggregate calibration hid an 85.2% miss rate on one malware family, revealing severe family-level reliability failures.

Predicting Vulnerability Attributes from CVE Descriptions

Mar 2026 – Aug 2026

- Trained models to predict vulnerability attributes from CVE security advisories using 2022–2023 CVEs, and evaluated on 36,539 CVEs from 2024.
- Reached 86.1% average accuracy across vulnerability-scoring components, and found that strong overall performance did not extend to rare and previously unseen labels.

PUBLICATIONS

- Investigating Usefulness of Security Threat Reports — manuscript under review
- Privacy and Paternalism: The Ethics of Student Data Collection — MIT Schwarzman College of Computing

LEADERSHIP

President, Women in Applied Cyber Club

Stanford University | 2025 – Present

- Led cybersecurity workshops and organized talks with industry professionals.

TECHNICAL SKILLS

Programming: Python, C++

ML: PyTorch, scikit-learn, LightGBM, Hugging Face Transformers, pandas, NumPy

Systems/Tools: Linux, Git, Bash, Docker